



Guidance for Repairing EM5.9 LED

ROE Technical Service Department

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1. Preparation

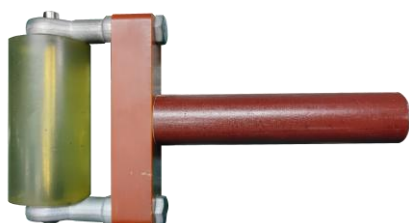
Tools



Cotton swab



Tweezers



Roller



Syringe



Electric screw machine



Soldering iron

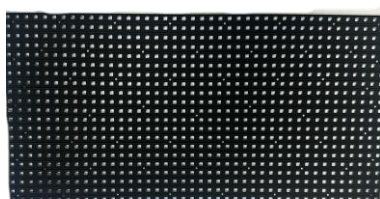


anti-static gloves

Materials



EM5.9 LED
(SAP: 205002S1320)



EM5.9 module mask
(SAP:304001-01948)



Tin wire



Glue

Storage temperature: $-20^{\circ}\text{C} \pm 5^{\circ}\text{C}$

Allow it to reach room temperature for 4 hours before use.

(SAP:310004-00447-01)



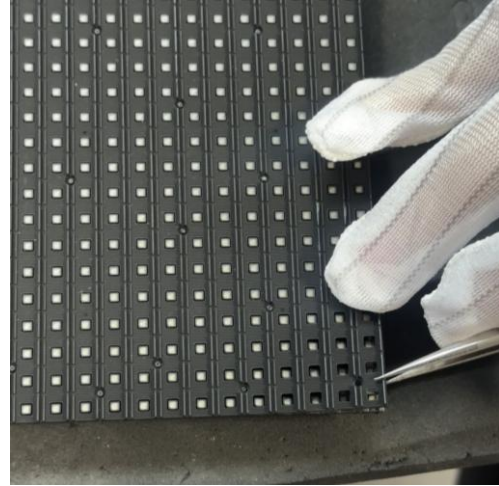
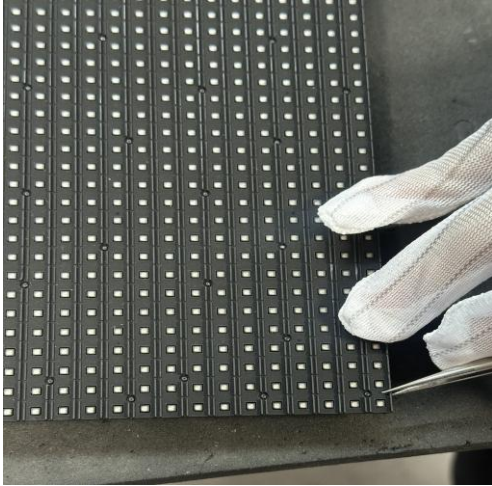
Industrial alcohol

2.Remove the bad LED

Take the module out from panel, put the module on the table.

Step1: Remove the mask.

Use tweezers to pry open the corner of the mask but make sure not to touch the LED.



Step2: Tear apart mask.

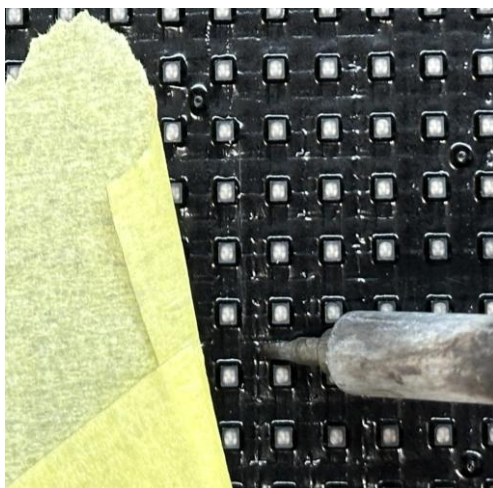
Once the mask is far enough from the module you can remove diagonally with your hand.



Do this slowly.

Step3: Clean the glue.

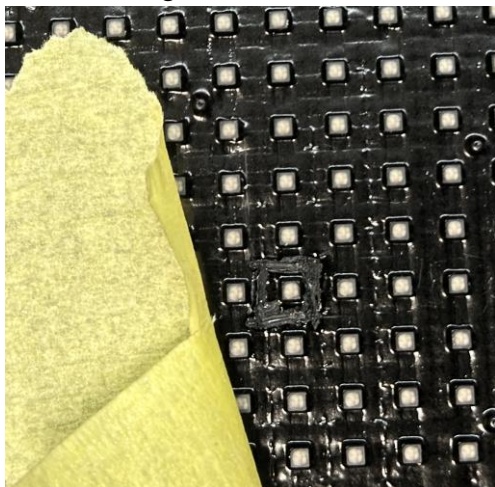
Locate the LED that needs repair, and use the soldering iron to clean the glue around the LED



**Be careful not to damage
the adjacent LEDs**

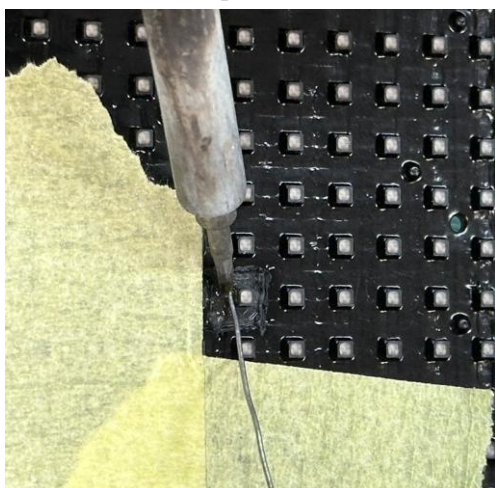
Step4: Remove the glue.

Using the soldering iron, trace a square around the glue surrounding the LED, then use tweezers to pinch and remove the glue.



Step5: Remove LED with soldering tip.

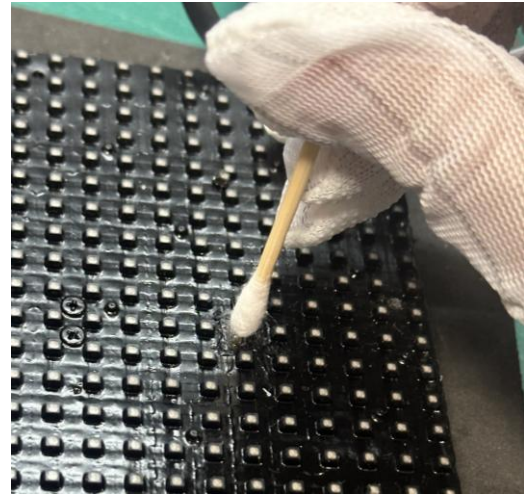
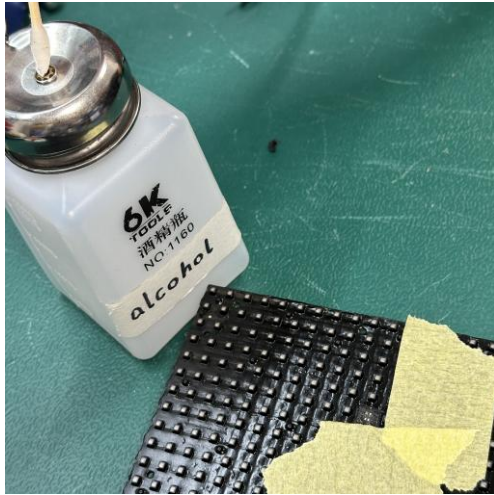
Using the soldering iron, push the LED up a little bit so that the contact points of the LED are no longer attached to the PCB points.



3. Weld a new LED

Step1: Clean the LED board.

Clean the solder residue on the LED board, and then use a cotton swab dipped in alcohol to clean the board. Be careful that the alcohol does not touch the other LED.

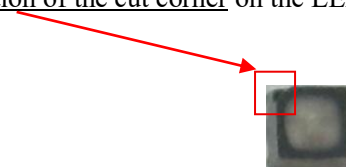


Step2: Place the new LED.

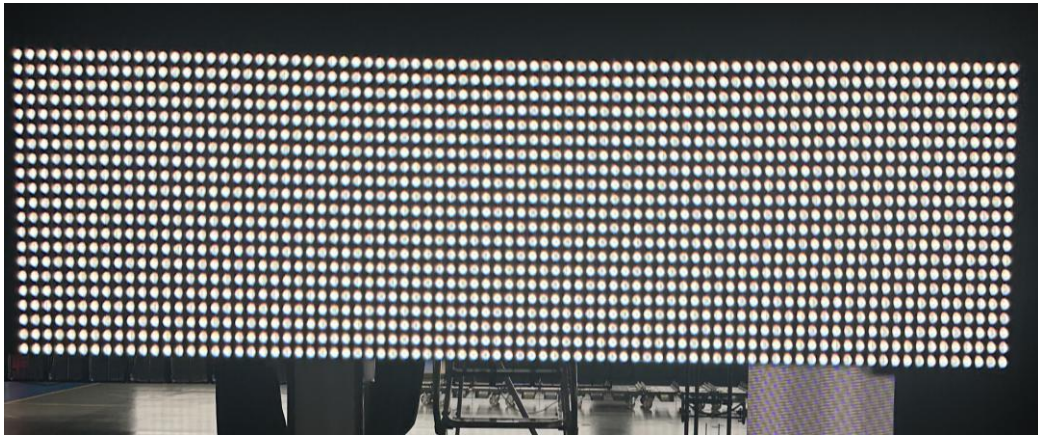
Solder the prepared LED of the same model onto the LED LED board.



Tip: Keep the orientation of the cut corner on the LED consistent with the LED on the module.

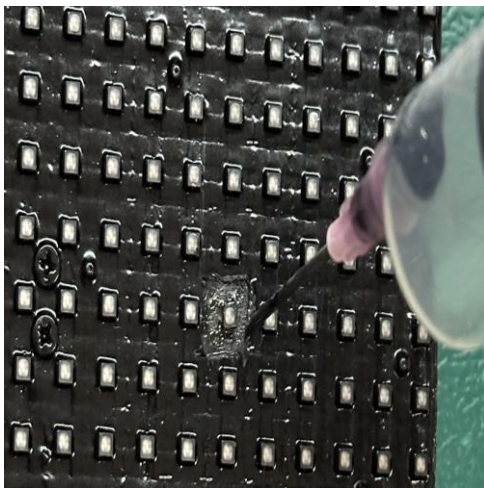


Step3: Test the module.



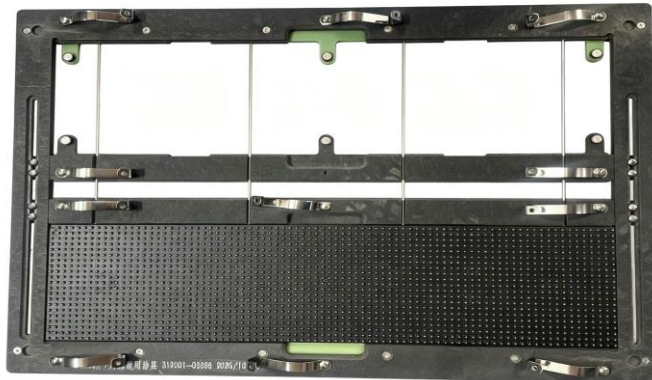
Step4: Add the glue.

Apply glue around the LED and let it dry under a fan for 120 minutes.



Tip: Ensure the glue applied on the LED board surface is smooth, and does not exceed the height of the original glue.

Step5: Place the module on the fixture.



**All masks must be removed
before baking**

Step6: Bake LED board.

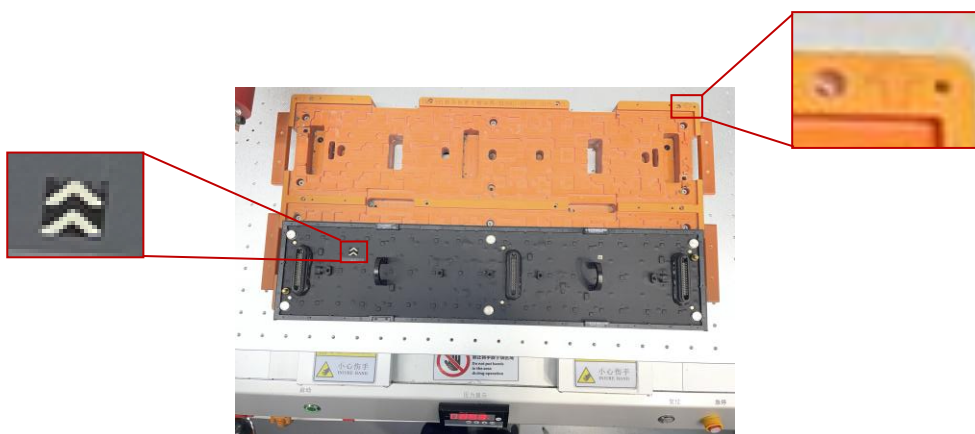
Bake modules at 125°C for 15 minutes after the oven reaches set temperature, with the exhaust fan set to level 3.



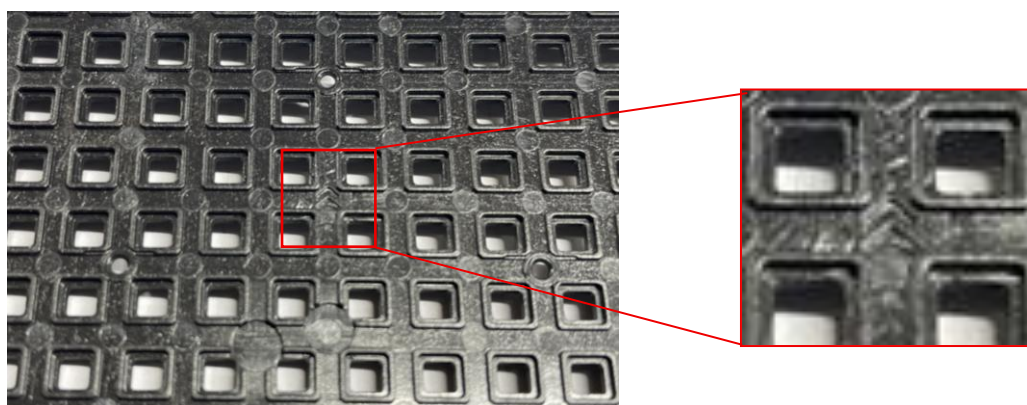
Remove the LED board from the oven and check whether the glue has fully cured. You can replace the new mask once the glue is fully cured.

Step7: Press the mask.

Place the LED board on the fixture and align the module direction with the fixture direction.

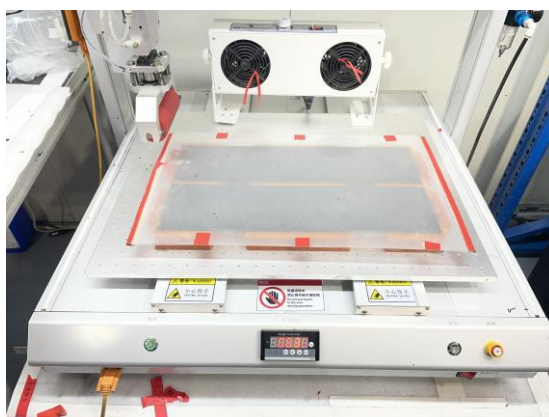


Ensure the LED side is facing up, and that the arrow direction on the back of the mask is consistent with the module direction before manually pre-pressing the four corner clips of the mask.



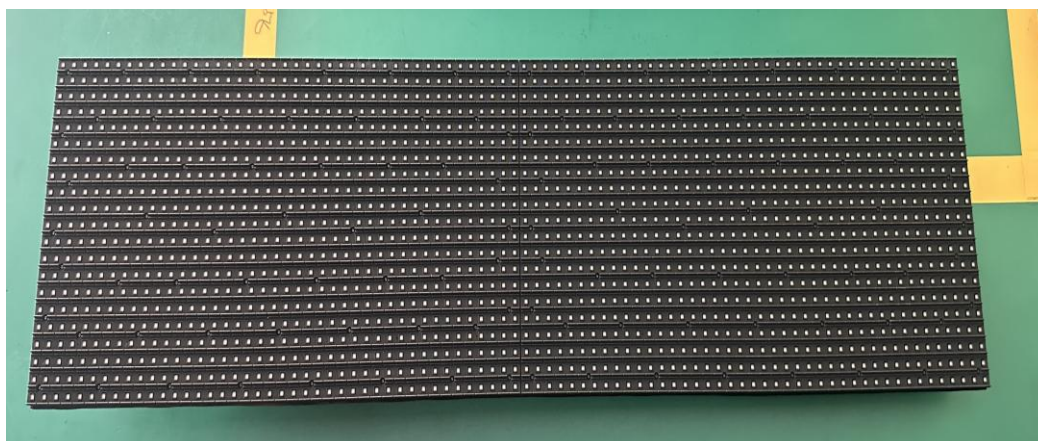


Cover the LED surface with PET film and use the mask-pressing machine to press the mask



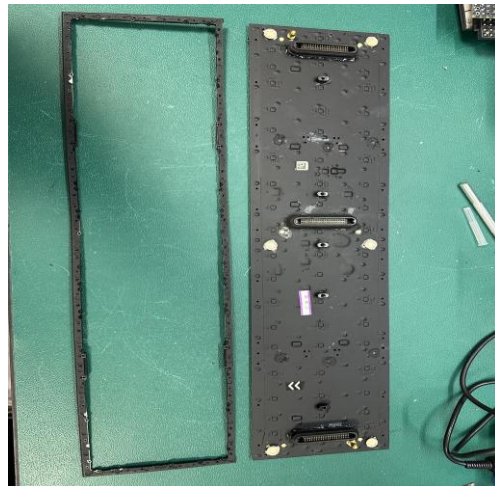
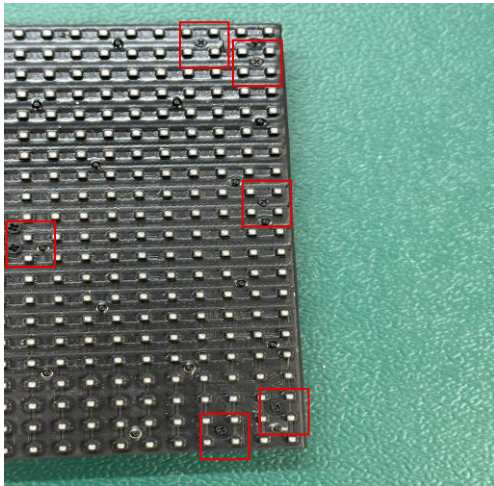
Can be replaced by a roller.

Final effect:



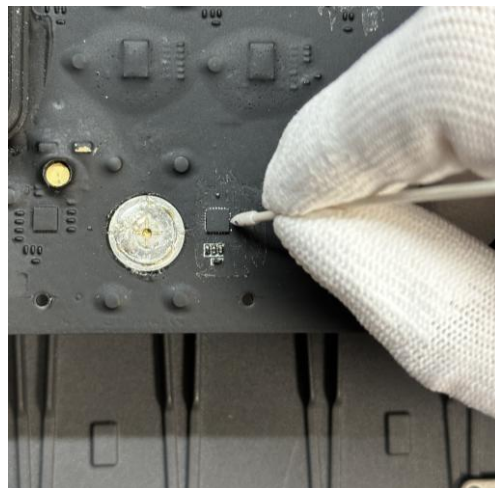
4.Remove the bad IC

Step1: Remove the LED panel bracket and adhesive strips.



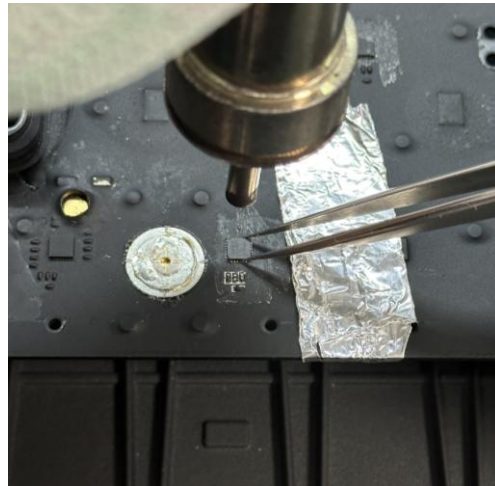
If the bad IC is located near the LED panel bracket and adhesive strips, this step is only needed.

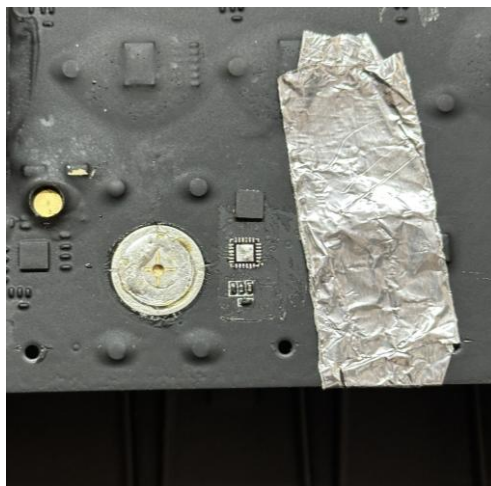
Step2: Clean the coating.



Step3: Remove the bad IC.

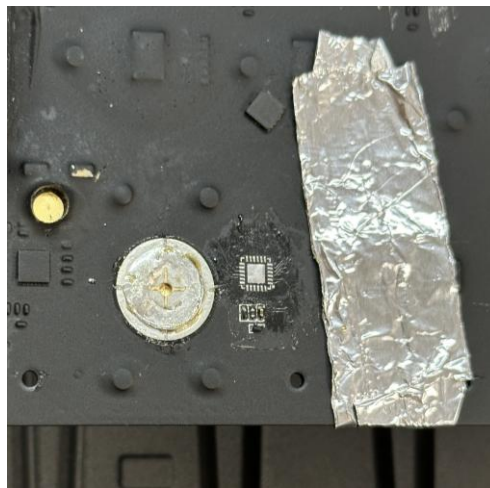
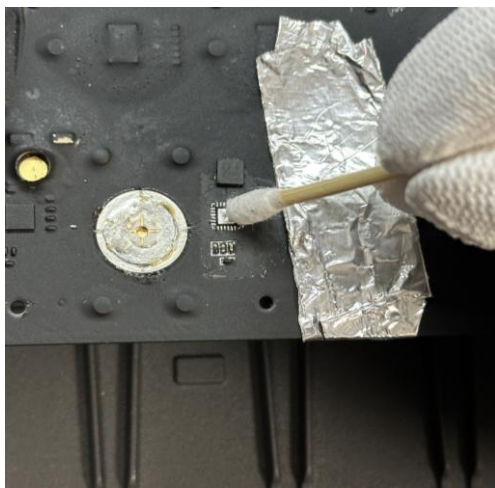
Set the hot-air gun to 350 °C, aim it at the IC and remove the faulty IC with tweezers.





5. Weld a new IC

Step1: Clean the solder pad.



Step2: Place the new IC

Use tweezers to pick up a new IC and place it on the solder pad. Set the hot-air gun to 350 °C, aim it at the IC, and use tweezers to align the IC with the pad and gently press it flush.



Step3: Test the module.

